Test Report

Weather Forecast Web App - https://weather-puce-six.vercel.app/

1. Summary

Total	Passed	Failed	Skipped	Pass rate
40	**30**	**10**	**0**	**75.0%**



10 known defect(s) confirmed (1 critical, 4 major, 5 minor) and 0 other failure(s) to investigate.

2. Test environment

Item	Detail
Application	https://weather-puce-six.vercel.app/
Browser	Edge (engine: MicrosoftEdge 154.0.4258.62)
Operating system	Windows-11-10.0.26200-SP0
Python / Selenium / pytest	Python 3.14.0 / Selenium 4.50.0 / pytest 9.1.1
Executed on	08-Oct-2026 08:28
Total run time	3.6 minutes
Test data	Tests marked live used the real site and services. All other tests used fixed data in place of the weather services.

3. Test results

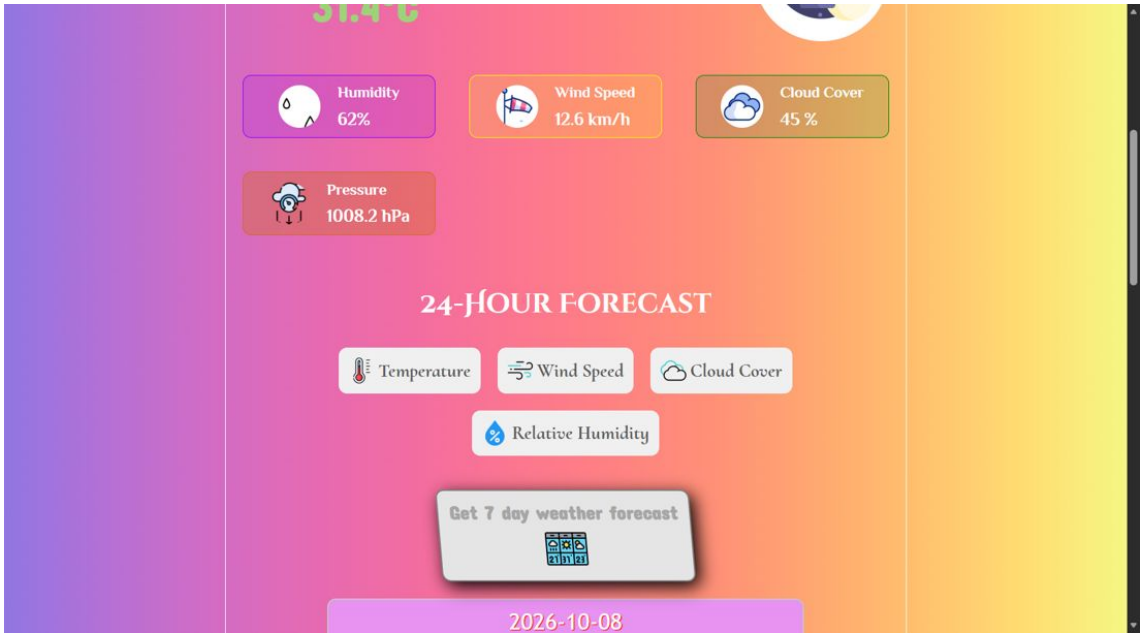
ID	Test scenario	Result	Time (s)	Notes
TC01	Home page loads	PASS	1.7	
TC02	Page load time	PASS	1.3	
TC03	Live search real city	PASS	5.0	
TC04	Search valid city shows weather	PASS	1.9	
TC05	Current conditions match api data	PASS	2.4	
TC06	Day night icon [1-day.png]	PASS	1.9	
TC06	Day night icon [0-night.png]	PASS	1.9	
TC07	Hourly buttons switch heading [temperature-temperature]	PASS	2.0	
TC07	Hourly buttons switch heading [wind-wind speed]	PASS	2.0	
TC07	Hourly buttons switch heading [cloud-cloud cover]	PASS	3.1	
TC07	Hourly buttons switch heading [humidity-relative humidity]	PASS	2.3	
TC08	Hourly temperature pagination	PASS	3.1	
TC09	Hourly values carry units [temperature-°C]	PASS	1.8	
TC09	Hourly values carry units [wind-km/h]	PASS	2.4	
TC09	Hourly values carry units [cloud-%]	PASS	2.2	
TC09	Hourly values carry units [humidity-%]	PASS	3.0	
TC10	Boundary buttons are disabled	FAIL	2.1	**D10** - Previous button should be disabled on the first page
TC11	Seven day forecast	PASS	2.4	
TC12	Daily min is not greater than max	FAIL	2.6	**D1** - 2026-10-08: Min shows 34.1 but Max shows 24.2
TC13	Sunrise and sunset format	PASS	1.8	
TC14	Empty search shows error	PASS	1.7	
TC15	Invalid city shows error [zzzzqq]	PASS	1.9	
TC15	Invalid city shows error [12345]	PASS	2.2	
TC15	Invalid city shows error [!!!@@@]	PASS	1.7	
TC15	Invalid city shows error [ ]	PASS	2.3	
TC16	App recovers after error	FAIL	9.9	**D2** - A valid search after an error does not show weather any more
TC17	Search history suggestions	PASS	2.3	
TC18	Search history survives reload	PASS	2.0	
TC19	Search history limited to five	PASS	1.7	

ID	Test scenario	Result	Time (s)	Notes
TC20	Enter key starts search	FAIL	6.9	**D3** - Pressing Enter in the search box does not start a search
TC21	Missing state is not shown as undefined	FAIL	1.9	**D4** - assert 'undefined' not in 'Singapore, undefined, SG'
TC22	Current location success	PASS	2.5	
TC23	Current location text has no trailing commas	FAIL	2.7	**D5** - Connaught Place, New Delhi, Delhi, India, ,
TC24	Location permission denied recovers	FAIL	7.0	**D6** - After location is denied the button stays on 'Searching..' and the search button stays hidden
TC25	Forecast api failure shows error	FAIL	6.6	**D7** - No error is shown to the user when the forecast service fails
TC26	Layout fits small screens [375]	PASS	1.8	
TC26	Layout fits small screens [768]	PASS	2.4	
TC27	No api key hardcoded in client script	FAIL	1.7	**D8** - An API key is hard-coded in source.js and visible to every visitor
TC28	Icon only buttons have accessible names	FAIL	1.4	**D9** - Buttons without an accessible name: ['search_button', 'stats_button', 'stats_button']
TC29	Script in search box is not executed	PASS	1.7	

4. Defects found

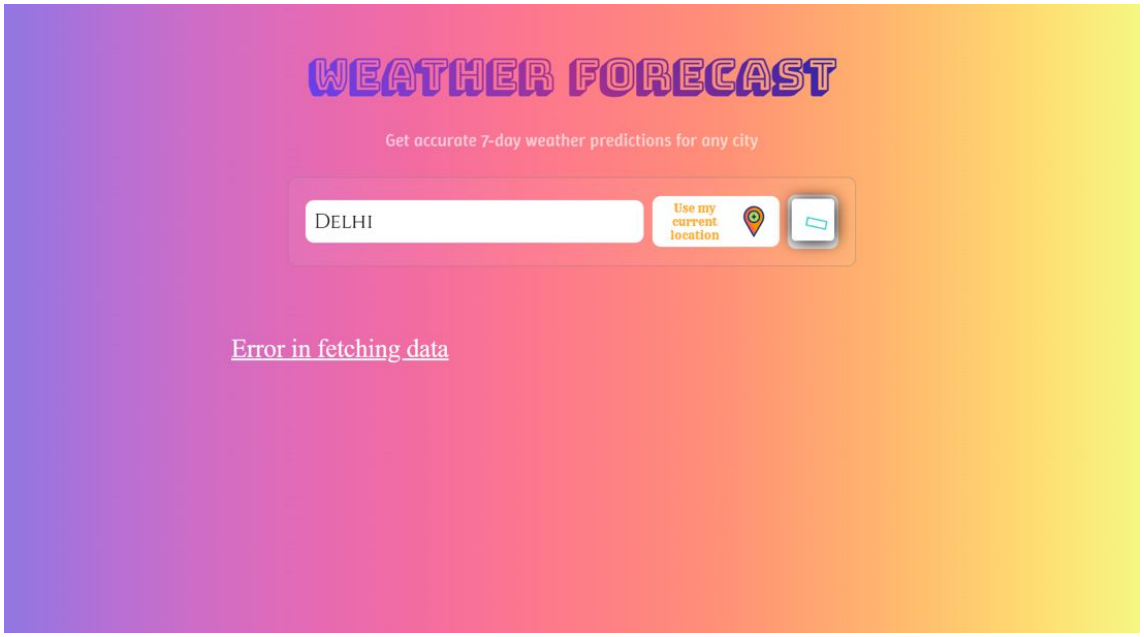
D1: In the 7-day cards, Min shows the higher temperature and Max the lower one

Severity	Major
Category	Data accuracy
Found by	TC12 - Daily min is not greater than max
Observed	2026-10-08: Min shows 34.1 but Max shows 24.2



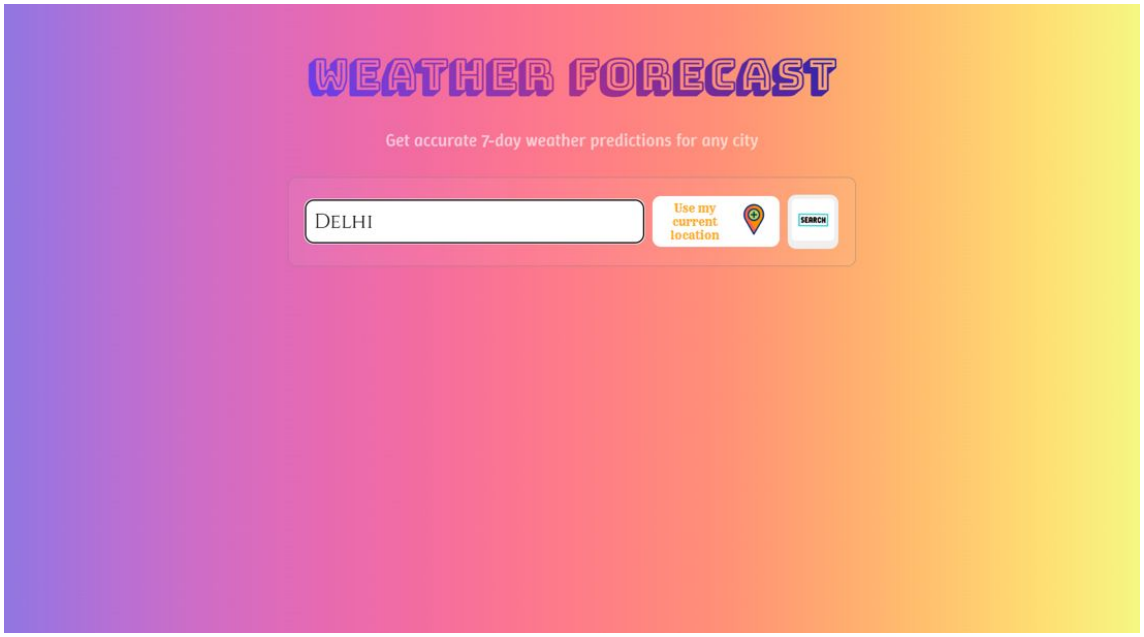
D2: After an error, a valid search never shows weather again until the page is reloaded

Severity	**Major**
Category	Functional
Found by	TC16 - App recovers after error
Observed	A valid search after an error does not show weather any more



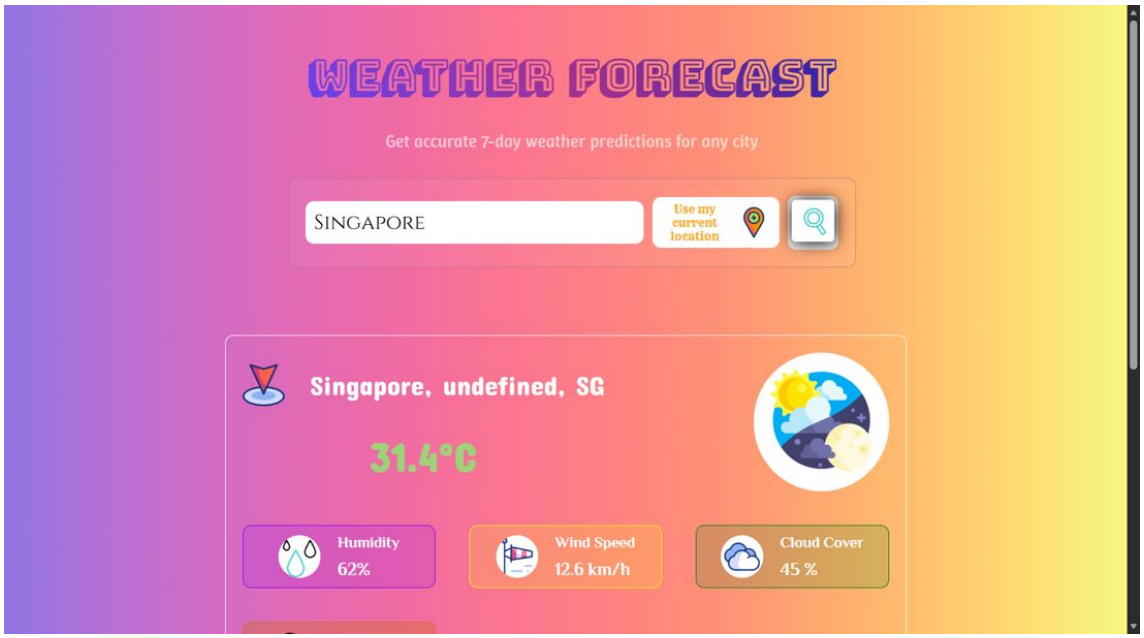
D3: Pressing Enter in the search box does not start a search

Severity	**Minor**
Category	UI / usability
Found by	TC20 - Enter key starts search
Observed	Pressing Enter in the search box does not start a search



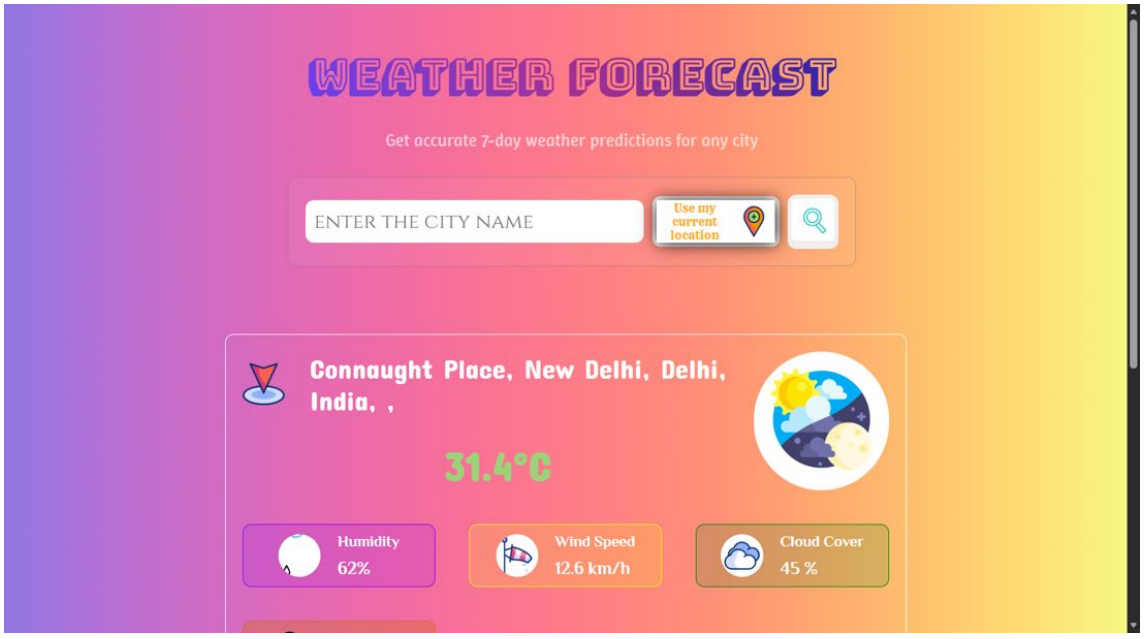
D4: A city with no state is shown with the word 'undefined' in its name

Severity	**Minor**
Category	Data display
Found by	TC21 - Missing state is not shown as undefined
Observed	assert 'undefined' not in 'Singapore, undefined, SG'



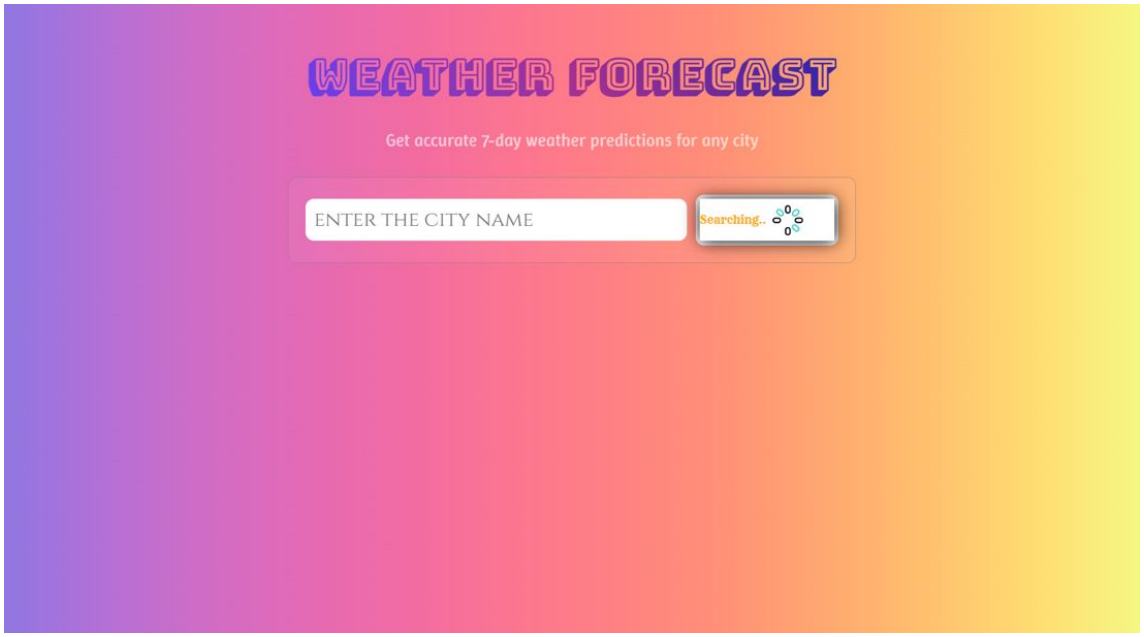
D5: The name from 'Use my current location' ends with ', ,'

Severity	**Minor**
Category	Data display
Found by	TC23 - Current location text has no trailing commas
Observed	Connaught Place, New Delhi, Delhi, India, ,



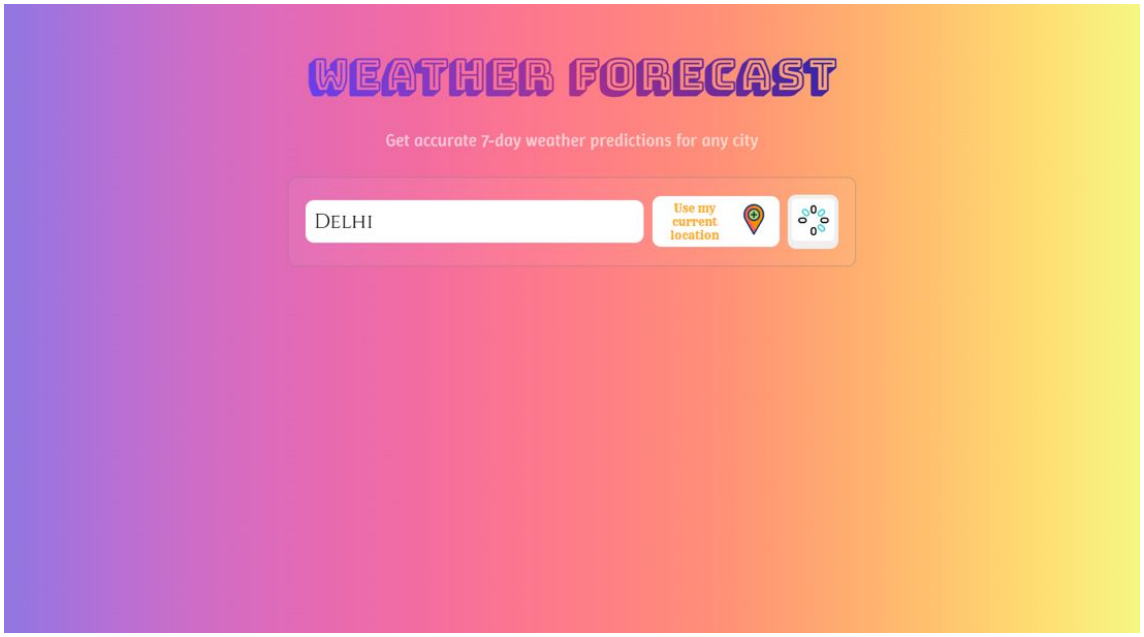
D6: When location permission is denied the button stays on 'Searching..' and the search button stays hidden

Severity	**Major**
Category	Error handling
Found by	TC24 - Location permission denied recovers
Observed	After location is denied the button stays on 'Searching..' and the search button stays hidden



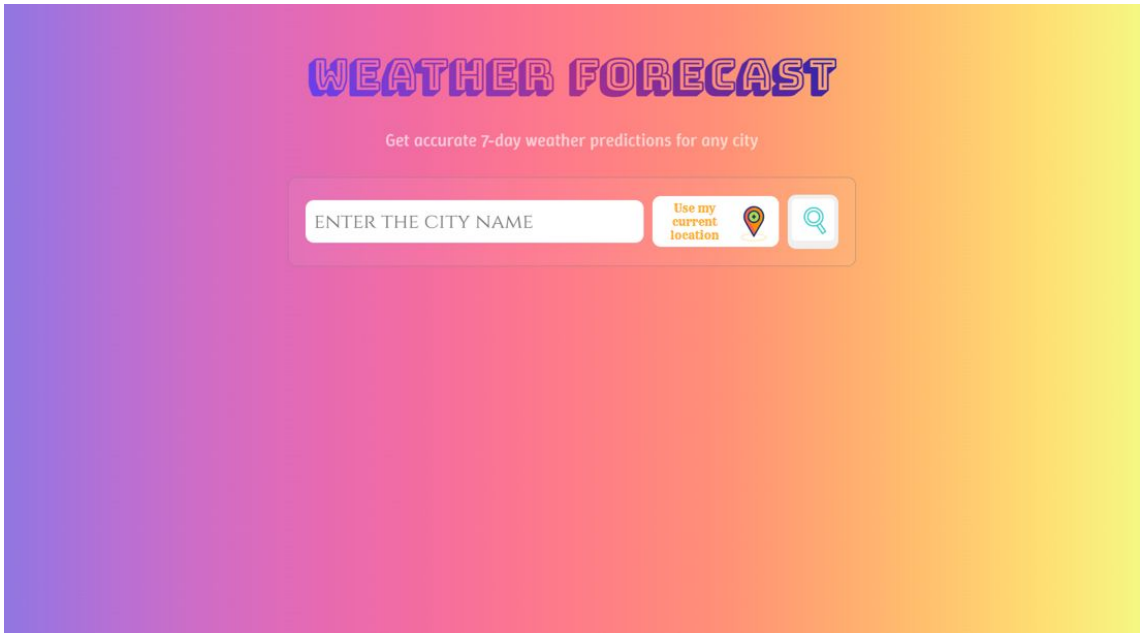
D7: No error is shown to the user when the forecast service fails

Severity	**Major**
Category	Error handling
Found by	TC25 - Forecast api failure shows error
Observed	No error is shown to the user when the forecast service fails



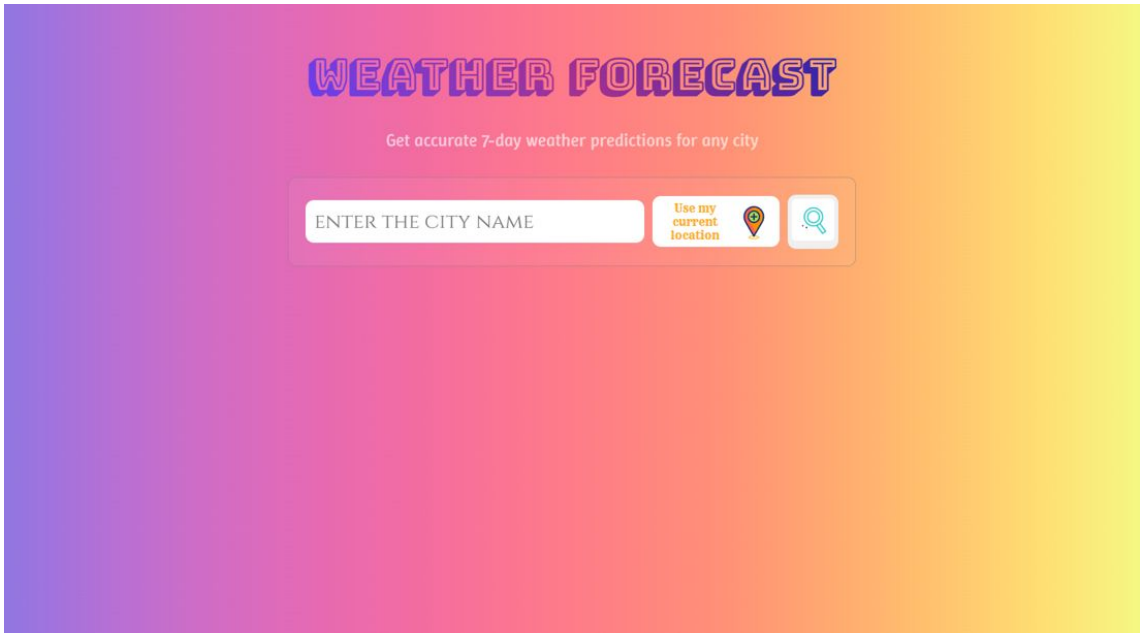
D8: An API key is written in source.js and visible to every visitor

Severity	**Critical**
Category	Security
Found by	TC27 - No api key hardcoded in client script
Observed	An API key is hard-coded in source.js and visible to every visitor



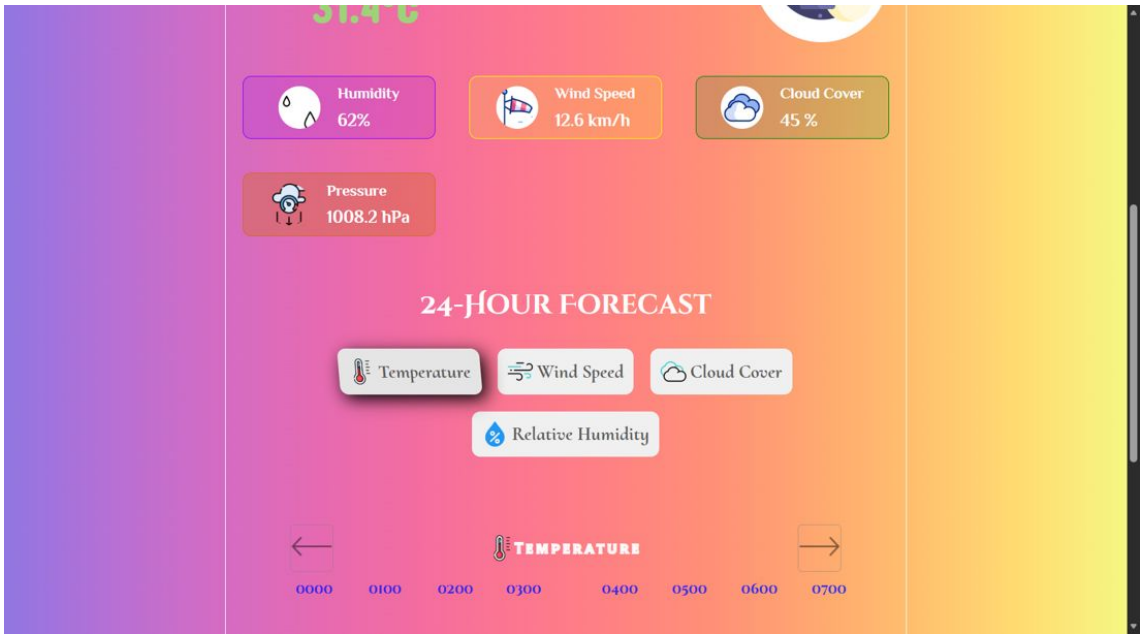
D9: Search, previous and next buttons have no accessible name

Severity	**Minor**
Category	Accessibility
Found by	TC28 - Icon only buttons have accessible names
Observed	Buttons without an accessible name: ['search_button', 'stats_button', 'stats_button']



D10: Previous and Next buttons never look disabled

Severity	**Minor**
Category	UI / usability
Found by	TC10 - Boundary buttons are disabled
Observed	Previous button should be disabled on the first page



5. Exploratory testing

Exploratory testing was organised in time-boxed sessions, each with a charter. The charters below guide the manual sessions. The observations list comes from inspecting the application code and behaviour; each one should be confirmed during a manual session.

Charter	Area	Ideas explored
E1	Search with unusual inputs	Very long names, accents, & and # characters, upper and lower case, repeated and quick searches
E2	Forecast screens	Compare values with another weather site, switch parameters quickly, search another city with the 7-day view open
E3	Location and permissions	Allow, block and allow again; desktop without GPS; offline mode
E4	Screen sizes and browsers	320, 375, 768 and 1366 px, zoom 150 percent, Edge and Brave
E5	Failure situations	Network off after load, slow connection, blocked services

Observations

#	Observation	Category	Severity
1	The '24-Hour Forecast' always starts at 00:00 of today instead of the current hour	Functional	Minor
2	Heading says 'Relative humidity' while the button says 'Relative Humidity'	UI / usability	Minor
3	Dates in the 7-day view appear as 2026-10-08 while the page design uses 27/08/2025	UI / usability	Minor
4	The city name is put in the request URL without encoding, so names with & or # may be cut short	Functional	Minor
5	Hourly button handlers are added again after every search	Code quality	Minor
6	'country' is used without being declared and 'findHistoryElem' is called but never defined	Code quality	Minor
7	The HTML has a head element inside body and a meta http-equiv tag with no content	Code quality	Minor
8	About 14 Google font families are loaded, which slows the first page load	Performance	Minor

6. Conclusion

The application is not ready for release: 30 of 40 tests passed (75.0%). Open defects: 1 critical, 4 major, 5 minor. Critical and major defects should be fixed and the regression suite repeated before release.